



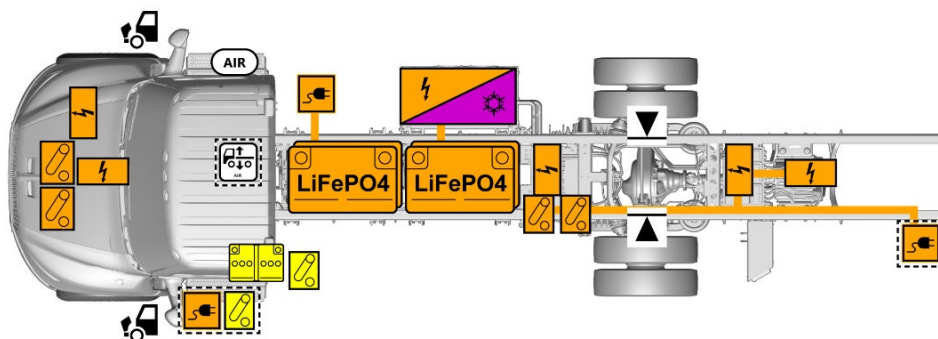
International® Electric MV™ Series Truck (eMV™) Model Year 2021–Present



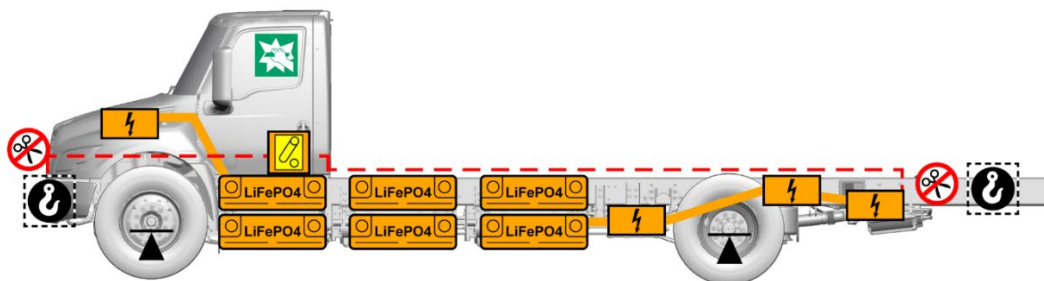
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690 V



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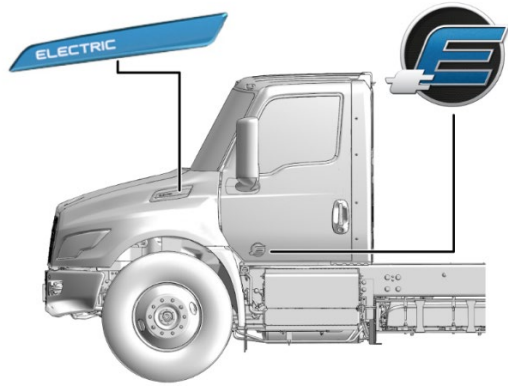


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WARNING: TO PREVENT PERSONAL INJURY AND / OR DEATH, NEVER CUT OR ATTEMPT TO BREACH HIGH-VOLTAGE BATTERIES, HIGH-VOLTAGE COMPONENTS, OR HIGH-VOLTAGE WIRING.

Low-Voltage Battery	High-Voltage Battery	High-Voltage Component	High-Voltage Charge Port	Battery Thermal Mgmt. System	Low-Voltage Disconnect
12V Shutoff for High-Voltage	High-Voltage Disconnect	Optional Tow Hook	High Voltage Power Cable	Lift Point	Compressed Air Tank
Break To Obtain Access	Hood Release	No Cut Below Line	Height Control Air	Optional Feature	Electric Propulsion

1. Identification / Recognition



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Figure 1. Exterior Decals and Badging

Exterior Identification

Badges and / or decals identifying vehicle as electric on door and hood.

Some localities may not include all badging or decals.



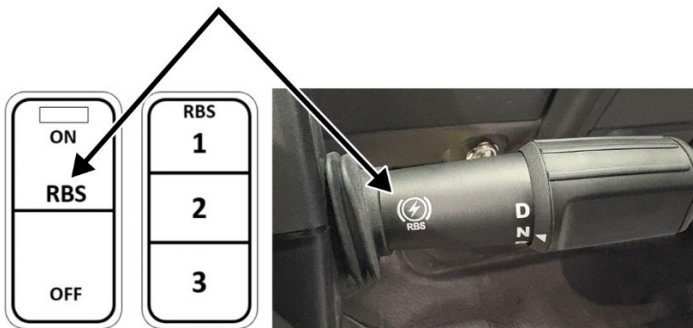
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Figure 2. Instrument Cluster

Interior Identification

1. REGEN
2. Regenerative Braking System (RBS) indicator, Levels 1, 2, 3

Regenerative Braking System (RBS) controls:
Dash mounted switches or steering column stalk (right-side of steering wheel).



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Figure 3. RBS

1. Identification / Recognition



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Figure 4. High-Voltage Cable and Component Labels



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Figure 5. High-Voltage Cables

High-Voltage Component Identification

High-voltage cables and components can be identified by the following:

- A triangle with a lightning bolt
- ORANGE in color and / or ORANGE with stripes

2. Immobilization / Stabilization / Lifting



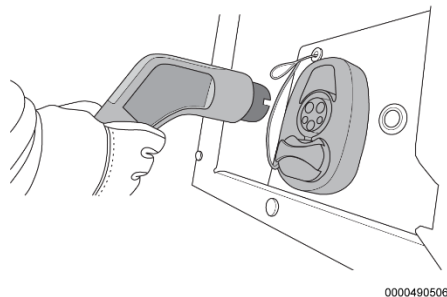
WARNING: TO PREVENT PERSONAL INJURY AND / OR DEATH, OR DAMAGE TO PROPERTY, DRAIN ALL AIR FROM THE SUSPENSION SYSTEM PRIOR TO MOVING A COMPROMISED VEHICLE WHENEVER POSSIBLE. FAILURE TO DO SO COULD RESULT IN THE VEHICLE UNCONTROLLABLY SHIFTING OR MOVING DUE TO AIR BEING RELEASED.



WARNING: TO PREVENT PERSONAL INJURY AND / OR DEATH, OR DAMAGE TO PROPERTY, DO NOT LIFT OR APPLY FORCE UNDER HIGH-VOLTAGE BATTERIES, COMPONENTS, OR CABLES. WHEEL LIFTS OR PLATFORM LIFTS ARE RECOMMENDED.



Use of an emergency plug inserted into the vehicle's charge port will not disconnect the high-voltage system.



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Figure 6. Emergency Plug

Install emergency plug in charge port.

2. Immobilization / Stabilization / Lifting

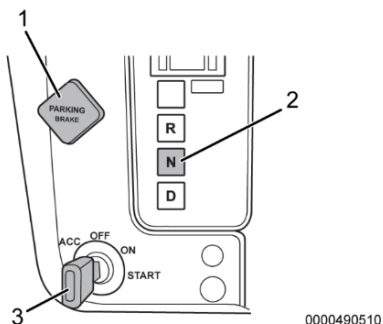


Figure 7. Dashboard



Figure 8. Stalk Shifter

1. Push Neutral (N) button (Figure 7, Item 2) or turn to Neutral (N) position (Figure 8, Item 1).
2. Set parking brake (Figure 7, Item 1).
3. Turn key OFF (Figure 7, Item 3).

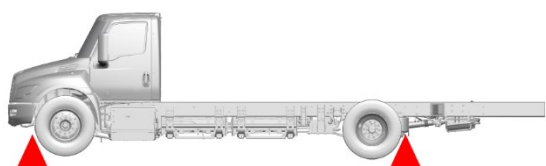


Figure 9. Wheel Chocks

Install wheel chocks.



Raise the vehicle under the axles at the lifting points shown. Install support stands under the axles and side members for stability.



Only use tow hooks to lift or recover the vehicle short distances.



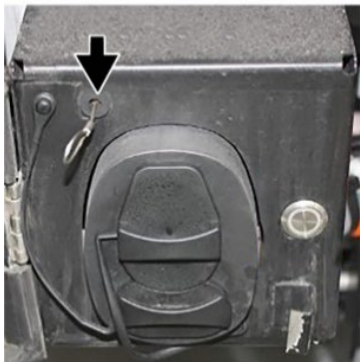
Figure 10. Switch

Lower the rear height of the vehicle by releasing the air in the rear air springs via the in-cab dash-mounted switch.

3. Disable Direct Hazards / Safety Regulations



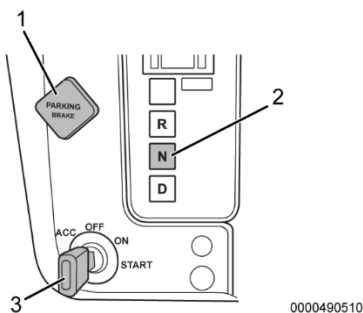
WARNING: TO PREVENT PERSONAL INJURY AND / OR DEATH, ALWAYS REMEMBER THE HIGH-VOLTAGE DISCONNECT SWITCH ONLY ISOLATES HIGH-VOLTAGE FROM COMPONENTS AFTER THE S-BOX. THE HIGH-VOLTAGE DISCONNECT SWITCH DOES NOT DISCONNECT HIGH-VOLTAGE IN THE S-BOX, THE HIGH-VOLTAGE BATTERIES, OR THE CABLES BETWEEN THESE ITEMS. THE VEHICLE'S 12 VOLT SYSTEM COULD STILL BE ENERGIZED. ALWAYS SHUT OFF BOTH THE HIGH-VOLTAGE DISCONNECT SWITCH AND THE 12V DISCONNECT SWITCH WHENEVER INTERACTING WITH A DAMAGED VEHICLE. NEVER TOUCH ORANGE CABLES WITHOUT WEARING APPROPRIATE HIGH-VOLTAGE PERSONAL PROTECTION EQUIPMENT (PPE).



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Figure 11. Emergency Charger Plug Release Cable

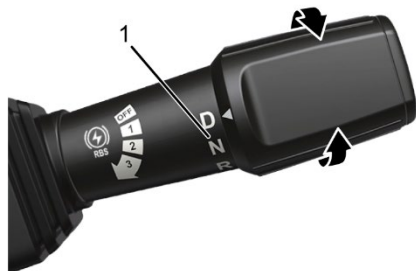
If necessary, pull emergency charger plug release cable and remove the charger from the port.



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Figure 12. Dashboard

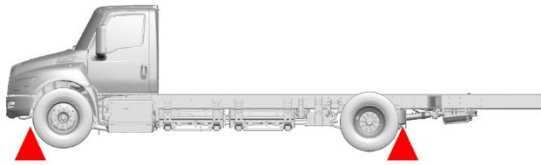
1. Push Neutral (N) button or turn to Neutral (N) position.
2. Set parking brake.
3. Turn key OFF.



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Figure 13. Stalk Shifter

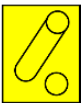
3. Disable Direct Hazards / Safety Regulations



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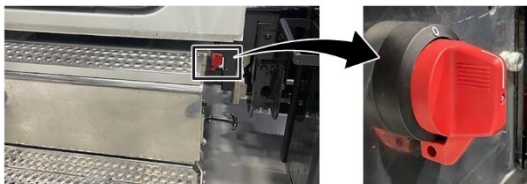
Figure 14. Wheel Chocks

Install wheel chocks.



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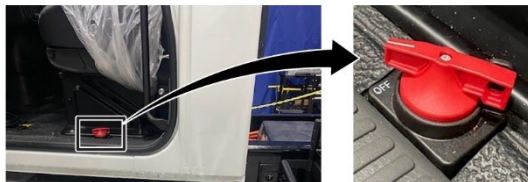
Figure 15. 12V Disconnect Switch



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Figure 16. 12V Disconnect Switch

Shut down low voltage is possible at two places (exterior at steps or driver-side cab).



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Figure 17. High-Voltage Disconnect Switch

The only 12V disconnect switch to shut down system high voltage is located on the interior driver-side floor.

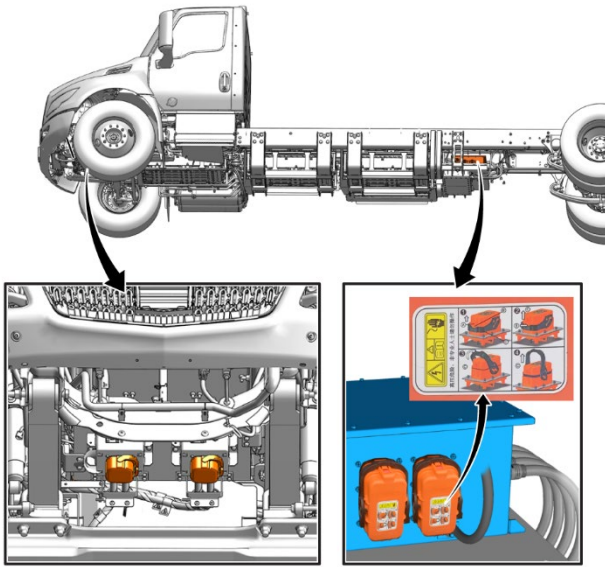


Wait 3 minutes for high-voltage energy to dissipate.



TO PREVENT PERSONAL INJURY AND / OR DEATH, HIGH-VOLTAGE COMPONENTS REMAIN LIVE AFTER THE HIGH-VOLTAGE DISCONNECT SWITCH IS TURNED OFF. THESE COMPONENTS ARE LOCATED PRIMARILY BETWEEN THE FRAME RAILS ON THE UNDERSIDE OF THE VEHICLE AND ARE LABELED.

3. Disable Direct Hazards / Safety Regulations



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Figure 18. Manual Service Disconnects (MSDs)

The two places to shut down component high voltage are located between the frame rails on the underside of the front of the vehicle and forward of rear axle.

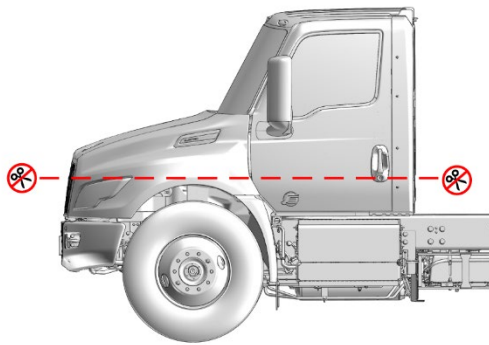
4. Access to the Occupants



The steering column will tilt and telescope. The driver seat can be adjusted by air-operated switches and / or mechanical lever.



The window is tempered glass and may be broken for access if needed.



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Figure 19. Do Not Cut Below Line

Do not cut below this line.

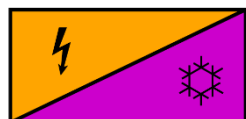
5. Stored Energy / Liquids / Gases / Solids



WARNING: TO PREVENT PERSONAL INJURY AND / OR DEATH, ALWAYS WEAR FULL PERSONAL PROTECTION EQUIPMENT (PPE), INCLUDING SELF-CONTAINED BREATHING APPARATUS (SCBA), WHENEVER FIRE IS INVOLVED.



- RED glycol-based automotive coolant



- RED glycol-based automotive coolant
- R134a refrigerant



- Air pressure of 100–140 psi (690–965 kPa)

6. In Case of Fire



WARNING: TO PREVENT PERSONAL INJURY AND / OR DEATH, ALWAYS WEAR FULL PERSONAL PROTECTION EQUIPMENT (PPE), INCLUDING SELF-CONTAINED BREATHING APPARATUS (SCBA), WHENEVER FIRE IS INVOLVED. FIRES IN CRASH-DAMAGED ELECTRIC VEHICLES COULD EMIT TOXIC OR COMBUSTIBLE GASES. SMALL AMOUNTS OF EYE, SKIN, OR LUNG IRRITANTS MAY BE PRESENT. IF EXPOSED, RINSE WITH LARGE AMOUNTS OF WATER FOR 10–15 MINUTES. CONSIDER THE ENTIRE VEHICLE AS ENERGIZED.



WARNING: TO PREVENT PERSONAL INJURY AND / OR DEATH, PAY ATTENTION TO SECONDARY FIRE EVENTS. THERE IS A HIGH RISK OF REIGNITION AFTER FIRE IS EXTINGUISHED.



7. In Case of Submersion



WARNING: TO PREVENT PERSONAL INJURY AND / OR DEATH, PAY ATTENTION TO SECONDARY FIRE EVENTS. THERE IS A HIGH RISK OF REIGNITION DUE TO DAMAGE AND CORROSION. SALTWATER INCREASES THIS RISK FOR ELECTRICAL SHORTS POST INCIDENT. KEEP FULL PERSONAL PROTECTION EQUIPMENT (PPE) READY, INCLUDING SELF-CONTAINED BREATHING APPARATUS (SCBA).



WARNING: TO PREVENT PERSONAL INJURY AND / OR DEATH, HANDLE A SUBMERGED VEHICLE WITH APPROPRIATE PERSONAL PROTECTION EQUIPMENT (PPE). CONSIDER THE ENTIRE VEHICLE AS ENERGIZED.



WARNING: TO PREVENT PERSONAL INJURY AND / OR DEATH, AVOID ANY CONTACT WITH A SUBMERGED HIGH-VOLTAGE SYSTEM. DO NOT ATTEMPT TO DISABLE THE HIGH-VOLTAGE SERVICE DISCONNECT SWITCH WHILE THE VEHICLE IS SUBMERGED. THE VEHICLE KEY MAY BE TURNED TO THE OFF POSITION.

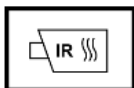


Follow Section 2 – Immobilization / Stabilization / Lifting and Section 3 – Disable Direct Hazards / Safety Regulations.

8. Towing / Transportation / Storage



WARNING: TO PREVENT PERSONAL INJURY AND / OR DEATH, PAY ATTENTION TO SECONDARY FIRE EVENTS. EVEN DAYS LATER SECONDARY FIRE EVENTS CANNOT BE EXCLUDED.



- Check battery temperature.

9. Important Additional Information

Accident assistance and vehicle recovery information for high-voltage systems can be found at:

- NFPA Alternative Fuel Vehicle Training
<http://www.nfpa.org/for-professionals/training-for-me/buy-alternative-fuel-vehicles-training>
- VDIK Accident Assistance and Recovery of Vehicles with High-Voltage or 48-Volt Systems
vdik.de/wp-content/uploads/2020/07/Accident_Assistance_Recovery_FAQ_en_072020_VDIK-1.pdf
- SAE J2990 Hybrid and EV First and Second Responder Recommended Practice
www.sae.org/standards/content/j2990
- National Transportation Safety Board (NTSB)
www.nts.gov

10. Explanation of Pictograms Used

Pictogram	Designation
	Electric Vehicle
	Battery, Low-Voltage
	Lithium Iron Phosphate Battery

10. Explanation of Pictograms Used



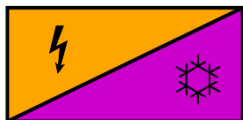
High-Voltage Component



Charge Port



Optional Charge Port



Battery Thermal Management System



Device to Shut Down Power in Vehicle



High-Voltage Disconnect



12V Shutoff for High-Voltage



Height Control Truck, by Air System

10. Explanation of Pictograms Used



Steering Wheel Tilt Control



Seat Height Adjustment by Air System



Seat Adjustment, Longitudinal



High-Voltage Power Cable



Air Tank



Break for Access



Warning, Electricity



General Warning Sign



Warning Low Temperature

10. Explanation of Pictograms Used



Use Thermal Infrared Camera



Use Water to Extinguish Fire



Wheel Chock



Use Only These Lifting Points



Hood Release



Optional Feature



Optional Tow Hook



No Cut Below Line



Explosive



Flammable

10. Explanation of Pictograms Used	
	Hazardous to Human Health
	Corrosives
	Acute Toxicity
	Environmental Hazard
	Gases Under Pressure